Catalogue of proofs

This list shows where to find the proofs in the notes. It does not fix a universal number of " required proofs": the groupings vary depending on the presentation.

Proved result	Proof
Comparing two sequences at infinity	p. 16
Increasing and not bounded above	p. 16
Bernoulli and geometric limits	p. 16
Limits of the exponential	p. 23
Powers and the exponential	p. 23
Convexity and tangents	p. 37
Limit of x ln(x) at zero	p. 52
Antiderivative given by an area, increasing case	p. 73
Integration by parts	p. 73
Pascal by counting	p. 30
Pascal by calculation	p. 30
Number of subsets: double counting	p. 30
Minimum distance to the plane	p. 59
Cartesian equation of a plane	p. 89
Probability of k successes	p. 44
Binomial expectation and variance	p. 81
Chebyshev on a finite sample space	p. 81
Derivative of the logarithm (differentiability assumed)	p. 52
Two antiderivatives differ by a constant	p. 67
Solutions of $y' = ay$	p. 68
Concentration of the mean, then the law of large numbers	p. 81